PERMANENCE PROPERTIES OF \'ETALE-SMOOTH LOCAL ON SOURCE-AND-TARGET PROPERTIES: \'ETALE LOCALITY ON THE SOURCE, SMOOTH LOCALITY ON THE TARGET, AND POSTCOMPOSITION/PERMANENCE FOR \'ETALE MORPHISMS

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author's opening sentence 'We write everything out completely.' is part of the counted proof text and is kept verbatim.

Editorial note. The counted span cites two definitions and two permanence lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- \'Etale-smooth local on source-and-target: definition. Stacks Project, Descent on Spaces, Definition definition-etale-smooth-local-source-target. Supplies parts (1), (2), (3) used to unfold and re-fold property P along coverings.
- Smooth local on source-and-target: definition. Stacks Project, Descent on Spaces, Definition definition-local-source-target. Supplies the target-locality unfolding used in parts (2) and (3).
- \'Etale permanence for algebraic spaces. Stacks Project, Properties of Spaces, Lemma spaces-properties-lemma-etale-permanence; Morphisms of Spaces, Lemma spaces-morphisms-lemma-etale-permanence. Supplies that X \to Y \times_Z X is \'etale in part (3) and that the section (f, 1) is \'etale in part (4).

Source reference. Original definition 0CFZ: sources/stacks-chow/tags.txt, source lines 15629–15629. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original definition 06F8: sources/stacks-chow/tags.txt, source lines 8242–8242. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 03FV: sources/stacks-chow/tags.txt, source lines 4506–4506. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05W3: sources/stacks-chow/tags.txt, source lines 7572–7572. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $\mathcal{P}$ be a property of morphisms of algebraic spaces over S which is étale-smooth local on source-and-target. Then*

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(1) $\mathcal{P}$ *is étale local on the source,*
(2) $\mathcal{P}$ *is smooth local on the target,*
(3) $\mathcal{P}$ *is stable under postcomposing with étale morphisms: if* $f : X \to Y$ *has* $\mathcal{P}$ *and* $g : Y \to Z$ *is étale, then* $g \circ f$ *has* $\mathcal{P}$*, and*
(4) $\mathcal{P}$ *has a permanence property: given* $f : X \to Y$ *and* $g : Y \to Z$ *étale such that* $g \circ f$ *has* $\mathcal{P}$*, then* f *has* $\mathcal{P}$*.*

Proof. We write everything out completely.

Proof of (1). Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\{X_i \to X\}_{i \in I}$ be an étale covering of X. If each composition $h_i : X_i \to Y$ has $\mathcal{P}$, then for each $|x| \in X$ we can find an $i \in I$ and a point $x_i \in |X_i|$ mapping to x. Then $(X_i, x_i) \to (X, x)$ is an étale morphism of pairs, and $\mathrm{id}_Y : Y \to Y$ is a smooth morphism, and h_i is as in part (3) of Definition 0CFZ. Thus we see that f has $\mathcal{P}$. Conversely, if f has $\mathcal{P}$ then each $X_i \to Y$ has $\mathcal{P}$ by Definition 0CFZ part (1).

Proof of (2). Let $f : X \to Y$ be a morphism of algebraic spaces over S. Let $\{Y_i \to Y\}_{i \in I}$ be a smooth covering of Y. Write $X_i = Y_i \times_Y X$ and $h_i : X_i \to Y_i$ for the base change of f. If each $h_i : X_i \to Y_i$ has $\mathcal{P}$, then for each $x \in |X|$ we pick an $i \in I$ and a point $x_i \in |X_i|$ mapping to x. Then $X_i \to X \times_Y Y_i$ is an étale morphism (because it is an isomorphism), $Y_i \to Y$ is smooth, and h_i is as in part (3) of Definition 06F8. Thus we see that f has $\mathcal{P}$. Conversely, if f has $\mathcal{P}$, then each $X_i \to Y_i$ has $\mathcal{P}$ by Definition 06F8 part (2).

Proof of (3). Assume $f : X \to Y$ has $\mathcal{P}$ and $g : Y \to Z$ is étale. The morphism $X \to Y \times_Z X$ is étale as a morphism between algebraic spaces étale over X (Properties of Spaces, Lemma 03FV). Also $Y \to Z$ is étale hence a smooth morphism. Thus the diagram

$$\begin{array}{ccc} X & \xrightarrow[f]{} & Y \\ \downarrow & & \downarrow \\ X & \xrightarrow{g \circ f} & Z \end{array}$$

works for every $x \in |X|$ in part (3) of Definition 06F8 and we conclude that $g \circ f$ has $\mathcal{P}$.

Proof of (4). Let $f : X \to Y$ be a morphism and $g : Y \to Z$ étale such that $g \circ f$ has $\mathcal{P}$. Then by Definition 0CFZ part (2) we see that $\mathrm{pr}_Y : Y \times_Z X \to Y$ has $\mathcal{P}$. But the morphism $(f, 1) : X \to Y \times_Z X$ is étale as a section to the étale projection $\mathrm{pr}_X : Y \times_Z X \to X$, see Morphisms of Spaces, Lemma 05W3. Hence $f = \mathrm{pr}_Y \circ (f, 1)$ has $\mathcal{P}$ by Definition 0CFZ part (1). □